

# The Catechol *ortho*-Cleavage Branch of the $\beta$ -Ketoadipate Pathway: A Mechanistic Review

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*A commissioned review-style synthesis of the CatA  $\rightarrow$  CatB  $\rightarrow$  CatC module that converts catechol to  $\beta$ -ketoadipate enol-lactone in aerobic bacteria.*

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## 1. Executive Summary

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The catechol *ortho*-cleavage branch is a compact, deeply conserved bacterial catabolic module that funnels a wide range of aromatic compounds into central metabolism. It accepts **catechol** — the convergence point of many "upper" aromatic pathways (benzoate via BenABCD, phenol, anthranilate, naphthalene, lignin-derived monomers) — and delivers  **$\beta$ -ketoadipate enol-lactone** to the shared "lower"  $\beta$ -ketoadipate reactions that feed the tricarboxylic acid (TCA) cycle. The module is executed by exactly three enzymes acting in an obligatory sequence: **CatA**, a catechol 1,2-dioxygenase that performs intradiol (*ortho*) ring cleavage to *cis,cis*-muconate; **CatB**, a muconate cycloisomerase (muconate lactonizing enzyme, MLE) that lactonizes *cis,cis*-muconate to (4S)-muconolactone; and **CatC**, a muconolactone  $\delta$ -isomerase that shifts the ring double bond to yield  $\beta$ -ketoadipate enol-lactone.

The striking feature of this three-step module — and the central mechanistic insight of this review — is that its three catalysts are drawn from **three structurally unrelated protein superfamilies** that are nonetheless organized into a single, coordinately regulated operon. CatA is a non-heme **Fe(III)** intradiol dioxygenase with a distinctive **2-His-2-Tyr** active site that is fundamentally different from the Fe(II)-2-His-1-carboxylate facial-triad chemistry of the *meta*-cleaving (extradiol) enzymes acting on the same substrate. CatB belongs to the **enolase superfamily** and uses a **Mn<sup>2+</sup>** ion (not the generic Mg<sup>2+</sup> of the broader superfamily) and a conserved lysine acid catalyst in an irregular ( $\beta/\alpha$ )<sub>7</sub> $\beta$ -barrel. CatC belongs to the small **muconolactone isomerase (MLI) family**, adopting a ferredoxin-type fold that assembles into a homodecamer whose active sites are built from residues contributed by two subunits.

Beyond the core enzymology, the branch is a model system for evolution and regulatory diversification. It is the demonstrated **evolutionary ancestor** of the plasmid-borne chlorocatechol ("modified *ortho*") pathways. The operon is typically induced by the pathway intermediate *cis,cis*-muconate acting through a LysR-type regulator (CatR in *Pseudomonas*), but both regulator identity (LysR vs. IclR) and gene order vary across lineages, illustrating that regulatory wiring is far more plastic than the conserved catalytic core. Finally, in *Pseudomonas putida* the catechol 1,2-dioxygenase step is elaborated by a second, non-redundant paralog (CatA2), which functions as a metabolic "safety valve" that resolves the toxic conflict arising when *ortho* and *meta* pathways are co-expressed. This review lays out the boundaries, mechanism, molecular players, evolutionary variation, physical constraints, and open questions of this system, drawing on primary structural, biochemical, genomic, and regulatory literature.

## 2. Definition and Biological Boundaries

### What is included

The catechol *ortho*-cleavage branch is defined here strictly as the **three-enzyme conversion of catechol to  $\beta$ -ketoadipate enol-lactone**:

| Step | Enzyme                                          | EC / role                           | Substrate $\rightarrow$ Product                                                   |
|------|-------------------------------------------------|-------------------------------------|-----------------------------------------------------------------------------------|
| 1    | <b>CatA</b> — catechol 1,2-dioxygenase          | EC 1.13.11.1; intradiol dioxygenase | catechol $\rightarrow$ <i>cis,cis</i> -muconate                                   |
| 2    | <b>CatB</b> — muconate cycloisomerase (MLE)     | EC 5.5.1.1; enolase superfamily     | <i>cis,cis</i> -muconate $\rightarrow$ (4S)-muconolactone                         |
| 3    | <b>CatC</b> — muconolactone $\delta$ -isomerase | EC 5.3.3.4; MLI family              | (4S)-muconolactone $\rightarrow$ $\beta$ -ketoadipate (3-oxoadipate) enol-lactone |

In *P. putida* KT2440 these correspond to the *cat* genes, with the exemplar loci **CatA-I/PP\_3713**, **CatA-II/PP\_3166**, **CatB/PP\_3715**, and **CatC/PP\_3714** [PMID: 12534466](#).

## Boundaries — what is upstream and downstream

The module **starts at catechol**, which is itself the output of diverse upper pathways: benzoate degradation via the *ben* operon (BenABCD), phenol hydroxylation, anthranilate 1,2-dioxygenation, and degradation of other aromatic hydrocarbons, amino aromatics, and lignin monomers [PMID: 8905091](#). The module **stops before** the shared "lower"  $\beta$ -ketoadipate reactions (the *pca*-encoded enol-lactone hydrolase,  $\beta$ -ketoadipate:succinyl-CoA transferase, and  $\beta$ -ketoadipyl-CoA thiolase) that convert the enol-lactone toward the TCA cycle. Importantly,  $\beta$ -ketoadipate enol-lactone is the point of **convergence with the parallel protocatechuate branch**: aromatic substrates degraded via protocatechuate (the *pca* genes) and via catechol (the *cat* genes) merge here and share the terminal enzymes [PMID: 8905091](#), [PMID: 16470664](#).

## What is often confused with it and should be treated separately

Several neighboring processes act on the *same substrate* (catechol) or produce *similarly named* intermediates and are frequently conflated:

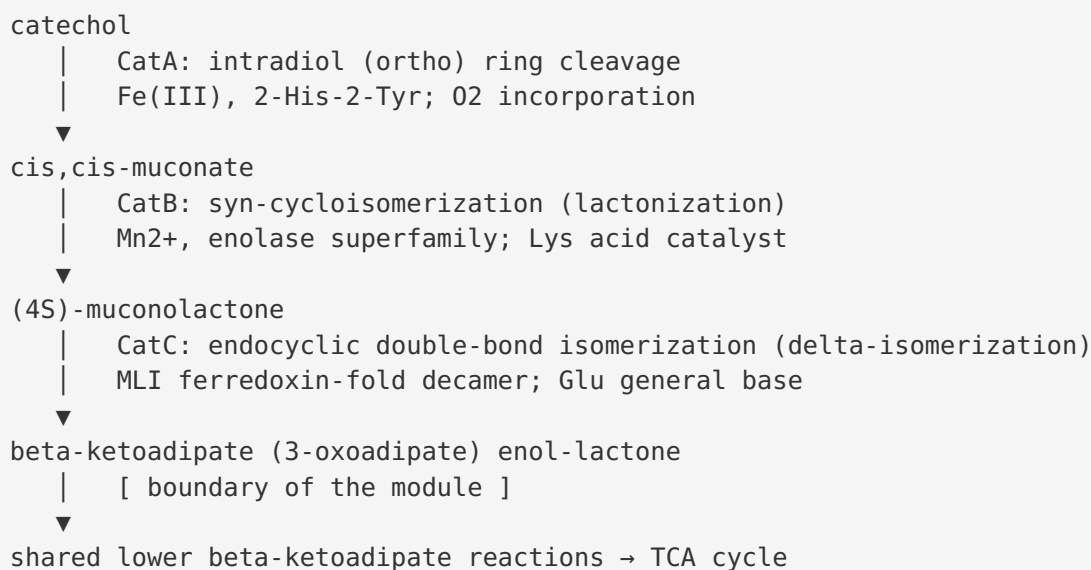
- **The *meta*-cleavage (extradiol) pathway.** Catechol 2,3-dioxygenase (e.g., metapyrocatechase from *P. putida* mt-2) also acts on catechol but cleaves the ring *extradiol* to 2-hydroxymuconate semialdehyde, using an Fe(II) center and an entirely different downstream route [PMID: 10368270](#). This is a competing, mutually exclusive fate for catechol — not part of the *ortho* branch.
- **The chlorocatechol "modified *ortho*" pathways.** Plasmid-borne *clc/tfd/cbn* operons process chlorosubstituted catechols with dedicated chlorocatechol 1,2-dioxygenases and chloromuconate cycloisomerases. Though homologous and evolutionarily derived from the *cat* enzymes, they are a separate module with distinct substrate specificities and downstream (dienelactone) chemistry [PMID: 8905091](#), [PMID: 7765840](#).
- **The protocatechuate branch.** Structurally analogous but not identical — protocatechuate 3,4-dioxygenase (an intradiol enzyme) opens protocatechuate, and its own cycloisomerase/isomerase steps feed the same  $\beta$ -ketoadipate convergence point. The two branches should be treated as parallel inputs, not the same branch.

## Competing definitions

The literature is largely consistent about the three-enzyme boundary, but usage of " $\beta$ -ketoadipate pathway" is broader in some reviews, where it denotes the entire funnel (both branches plus the shared lower steps). This review adopts the narrower, mechanistically coherent definition: the catechol branch = CatA + CatB + CatC.

### 3. Mechanistic Overview

The best-supported model is a strictly ordered, three-reaction relay in which each product is the obligatory substrate of the next enzyme:



**Step 1 — CatA (obligatory, committed, O<sub>2</sub>-dependent).** CatA performs intradiol cleavage: it inserts both atoms of molecular oxygen and breaks the C1–C2 bond *between* the two hydroxyls, yielding the aliphatic diacid *cis,cis*-muconate. This is the committed, ring-opening step and the branch point that distinguishes *ortho* from *meta* catabolism. Mechanistically, intradiol (Fe<sup>3+</sup>) and extradiol (Fe<sup>2+</sup>) dioxygenases converge on a **common proximal hydroperoxide intermediate**, but process it via **different Criegee 1,2-rearrangements**; notably, intradiol cleavage is pH-independent in its rate-determining step, indicating no acid–base catalysis there [PMID: 19300822](#).

**Step 2 — CatB (obligatory).** CatB catalyzes syn-cycloisomerization: an intramolecular Michael-type addition that closes the ring to form the five-membered lactone (4S)-muconolactone. As an enolase-superfamily member, it initiates catalysis by metal-assisted abstraction of the substrate α-proton to form an enolic intermediate, then directs that intermediate specifically toward cyclization [PMID: 8987982](#), [PMID: 19220063](#).

**Step 3 — CatC (obligatory).** CatC performs an endocyclic rearrangement of the double bond within the lactone ring, converting muconolactone to β-ketoadipate enol-lactone — the module's output [PMID: 23376735](#).

**Obligatory vs. conditional vs. accessory.** All three catalytic steps are **obligatory** — no catechol carbon reaches the lower pathway without passing through each in order. The **CatA2 paralog** in *P. putida* is **accessory/conditional**: it is dispensable under many conditions but becomes critical when catechol accumulates during simultaneous *ortho/meta* expression (Section 5). Regulatory events (CatR-mediated induction) are **conditional** — necessary for expression but not part of the chemical transformation.

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## 4. Major Molecular Players and Active Assemblies

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A defining, and perhaps counterintuitive, property of this module is that **three unrelated folds are strung together into one funnel**. Nature did not evolve a single superfamily to run all three steps; instead it recruited catalysts from three independent lineages.

### 4.1 CatA — catechol 1,2-dioxygenase (intradiol, Fe(III))

CatA is a **homodimer with one catalytic non-heme ferric ion per monomer**. The 1.8 Å crystal structure of the *Acinetobacter* enzyme revealed a "rudimentary design" — a five-helix N-terminal **helical zipper** forming the dimer interface, and, unexpectedly, **two phospholipids bound within an ~8 × 35 Å hydrophobic tunnel** along the dimer axis [PMID: 10801478](#).

The active-site chemistry is the key discriminator between *ortho* and *meta* enzymes. Work on the homologous intradiol enzyme **protocatechuate 3,4-dioxygenase (3,4-PCD)** from *P. putida* established that the active-site Fe(III) is ligated by an **axial Tyr447/His462 and equatorial Tyr408/His460/OH<sup>-</sup>** — a **2-His-2-Tyr** arrangement. When the catecholate substrate chelates the iron, the **axial tyrosinate and the hydroxide are displaced and act as in-situ bases** to deprotonate the substrate, priming it for O<sub>2</sub> attack [PMID: 16101286](#). This contrasts sharply with the **2-His-1-carboxylate facial triad** that coordinates Fe(II) in extradiol and many other mononuclear non-heme oxygenases [PMID: 9667935](#). Thus two enzyme classes act on the same substrate through structurally and mechanistically distinct iron sites — Fe<sup>3+</sup>/tyrosinate for *ortho*, Fe<sup>2+</sup>/facial-triad for *meta*. Complementary crystallographic and resonance-Raman studies of 3,4-PCD show the axial tyrosinate dissociating during turnover and returning during product release, defining new reaction-cycle intermediates [PMID: 9485360](#).

## 4.2 CatB — muconate cycloisomerase / muconate lactonizing enzyme (MLE)

CatB is an **enolase-superfamily** enzyme built on an irregular **( $\beta/\alpha$ )<sub>7</sub> $\beta$ -barrel** (only 7, not 8,  $\alpha$ -helices). The refined 1.85 Å structure of MLE from *P. putida* PRS2000 shows a **homo-octamer** of 373-residue subunits, each with an  $\alpha$ + $\beta$  capping domain, the barrel domain, and a C-terminal subdomain [PMID: 7500361](#). A key correction that emerged during this investigation concerns the metal: while the enolase superfamily is generically described as requiring a divalent cation (often stated as  $\text{Mg}^{2+}$ ), the canonical  $\beta$ -ketoadipate-pathway CatB uses an **essential  $\text{Mn}^{2+}$**  in the active-site cleft [PMID: 7500361](#), independently confirmed by the earlier 3 Å structure that located "the manganese ion that is essential for catalytic activity" [PMID: 3612800](#).

Catalysis proceeds by the superfamily signature — metal-assisted abstraction of the substrate  $\alpha$ -proton to generate an enolate intermediate — with a **conserved Lys at the end of the second  $\beta$ -strand of the barrel serving as the acid catalyst**. Remarkably, two divergent families achieve muconate cycloisomerization with opposite stereochemistry: the canonical family catalyzes **syn-cycloisomerization** to (4*S*)-muconolactone, while a second family catalyzes **anti-cycloisomerization** — two independent evolutionary solutions to the same net reaction [PMID: 19220063](#). The unifying enolase-superfamily principle —  $\alpha$ -proton abstraction to form an enolic intermediate that then partitions toward distinct products (racemization, elimination, cycloisomerization) — places CatB firmly in this mechanistically defined group [PMID: 8987982](#).

## 4.3 CatC — muconolactone $\delta$ -isomerase (MLI family)

CatC is a small protein adopting a **ferredoxin-type fold** and assembling into a **homodecamer of 52-symmetry** (a pentamer of dimers). Structural insight comes from the closely homologous chloromuconolactone dehalogenase **ClcF** from *Rhodococcus opacus* 1CP: the MLI-family active site is formed **at the interface of two monomers**, and a **conserved glutamate (E27) acts as the proton acceptor / general base** driving the double-bond isomerization; the authors explicitly note the conserved active-site architecture is "representative for the MLI family" (EC 5.3.3.4) [PMID: 23421784](#). The reaction itself — endocyclic rearrangement of the muconolactone double bond to give 3-oxoadipate enol-lactone — is well defined [PMID: 23376735](#). Mutational analysis of ClcF corroborates the role of E27 in initial proton abstraction and of neighboring residues in substrate binding [PMID: 22842338](#).

## Summary table of the three catalysts

| Feature                    | CatA                           | CatB                                                                 | CatC                           |
|----------------------------|--------------------------------|----------------------------------------------------------------------|--------------------------------|
| Reaction                   | Intradiol ring cleavage        | <i>syn</i> -cycloisomerization                                       | $\delta$ -isomerization        |
| Superfamily / fold         | Intradiol dioxygenase          | Enolase superfamily; ( $\beta/\alpha$ ) <sub>7</sub> $\beta$ -barrel | MLI family; ferredoxin fold    |
| Cofactor                   | Non-heme <b>Fe(III)</b>        | <b>Mn<sup>2+</sup></b>                                               | None (Glu general base)        |
| Active-site signature      | 2-His-2-Tyr; tyrosinate base   | Lys acid catalyst                                                    | Glu27 proton acceptor          |
| Oligomeric state           | Homodimer                      | Homo-octamer                                                         | Homodecamer (52 symmetry)      |
| O <sub>2</sub> -dependent? | Yes                            | No                                                                   | No                             |
| Key reference              | <a href="#">PMID: 10801478</a> | <a href="#">PMID: 7500361</a>                                        | <a href="#">PMID: 23421784</a> |

The three enzymes share no common fold, no common cofactor, and no common oligomeric state — the module is a functional assembly of independently evolved parts, unified only by substrate channeling through the metabolite relay and by co-regulation.

## 5. Evolutionary and Cell-Biological Variation

### 5.1 Regulatory wiring is plastic; catalysis is conserved

The *cat* operon is **inducibly controlled by the pathway intermediate *cis,cis*-muconate**. In *P. putida* the *catBCA* operon is activated by the LysR-type regulator **CatR** with *cis,cis*-muconate as the effector [PMID: 9858745](#); an internal CatR binding site within the *catB* gene provides a repressive, DNA-looping layer of fine control [PMID: 9573187](#). However, the **regulator identity and gene order are not conserved**:



| Organism                        | Gene arrangement            | Regulator type   | Reference                      |
|---------------------------------|-----------------------------|------------------|--------------------------------|
| <i>P. putida</i>                | <i>catR</i> / <i>catBCA</i> | LysR (CatR)      | <a href="#">PMID: 9858745</a>  |
| <i>Rhodococcus erythropolis</i> | <i>catRABC</i>              | <b>IclR-type</b> | <a href="#">PMID: 17483937</a> |
| <i>Halomonas organivorans</i>   | <i>catRBCA</i>              | LysR (CatR)      | <a href="#">PMID: 21695219</a> |

This variation — a LysR regulator in some lineages, an entirely unrelated IclR-type in others, with reshuffled gene order — demonstrates that the **regulatory layer is far more evolutionarily labile than the enzymatic core**. The chemistry is fixed; the control logic is repeatedly re-invented.

## 5.2 The branch is the ancestor of chlorocatechol pathways

The catechol branch is explicitly identified as the **evolutionary precursor** of the plasmid-borne *ortho*-pathways for chlorocatechol degradation [PMID: 8905091](#). At the sequence level, chlorocatechol 1,2-dioxygenases and chloromuconate cycloisomerases are clearly similar to their catechol counterparts, whereas the downstream dienelactone hydrolases are more distantly related and were likely recruited from elsewhere [PMID: 7765840](#). The regulatory homology extends too: ClcR (chlorocatechol) and CatR (catechol) are paralogous LysR regulators responding to the analogous inducers 2-chloro-*cis,cis*-muconate and *cis,cis*-muconate, respectively [PMID: 9858745](#), and their activation mechanisms appear evolutionarily conserved [PMID: 9171413](#). This lineage relationship is a textbook example of a chromosomal catabolic module serving as the toolkit from which specialized xenobiotic-degrading operons were later assembled.

## 5.3 Paralog elaboration: the CatA2 "safety valve"

The catechol 1,2-dioxygenase step, uniquely, has been elaborated by gene duplication. In *P. putida* mt-2 (a KT2440-background strain carrying the TOL plasmid pWW0), a **second chromosomal *catA* copy — *catA2*, located downstream of the *ben* operon — encodes an additional catechol 1,2-dioxygenase** [PMID: 24341396](#). These paralogs are **not redundant**. Knockout analyses show that **CatA2 counteracts the toxicity of the catechol surplus** generated when benzoate induces the *ortho* pathway, and thereby stabilizes the coexistence of the plasmid-borne *meta* (*xyl*/TOL) and chromosomal *ortho* (*cat*) pathways. Cells carrying pWW0 but lacking *catA2* lost the plasmid on benzoate and grew more slowly on benzyl



alcohol. CatA2 is best understood as a **metabolic safety valve** that resolves the conflict of co-expressed *ortho* and *meta* routes [PMID: 24341396](#). This is the strongest available evidence that the two CatA paralogs (PP\_3713 and PP\_3166) have functionally diverged.

## 5.4 Physiological and environmental variation

Additional variation is documented at the physiological level. Carbon catabolite repression modulates the pathway: in *P. putida*, addition of succinate to phenol-grown cells immediately represses translation of the Cat proteins (catechol 1,2-dioxygenase, muconate cycloisomerase, muconolactone isomerase), while the shared Pca (lower-pathway) enzymes are unaffected [PMID: 16478453](#). Enzyme thermostability also varies with ecological niche: a cold-adapted *Pseudomonas* sp. produces a thermolabile CatA active from 5–37 °C but rapidly inactivated above 25 °C, unlike the robust mesophilic *P. putida* mt-2 enzyme [PMID: 23720227](#).

## 6. Constraints, Dependencies, and Failure Modes

**Obligatory ordering.** The chemistry itself imposes the sequence: CatA's product *cis,cis*-muconate is the only substrate CatB can lactonize, and CatB's product (4S)-muconolactone is the only substrate CatC can isomerize. There is no bypass within the module — the order CatA → CatB → CatC is enforced by substrate identity, not merely by convention.

**Mutually exclusive fates of catechol.** Intradiol (*ortho*, CatA) and extradiol (*meta*, catechol 2,3-dioxygenase) cleavage are competing, mutually exclusive fates for the same molecule. The former uses Fe(III)/tyrosinate chemistry and leads to muconate; the latter uses Fe(II)/facial-triad chemistry and leads to hydroxymuconate semialdehyde [PMID: 10368270](#), [PMID: 19300822](#). When both routes are simultaneously active (e.g., chromosomal *ortho* + plasmid *meta*), the result is metabolic conflict and catechol toxicity — the very problem the CatA2 safety valve exists to solve [PMID: 24341396](#).

**O<sub>2</sub> dependency.** Only the first step consumes O<sub>2</sub> (both atoms are incorporated into muconate). The entire *ortho* branch is therefore strictly aerobic — a hard environmental constraint absent from the later isomerization steps.

**Toxic-intermediate failure mode.** Catechol and its oxidation products are cytotoxic. Accumulation — from CatA insufficiency, pathway imbalance, or auto-oxidation of related metabolites (e.g., naphthoquinone accumulation in a poorly balanced naphthalene degrader)

— causes growth inhibition and loss of viability [PMID: 17975081](#). This makes CatA flux capacity, not the downstream isomerases, the typical bottleneck and the point of paralog reinforcement.

**Metal dependency.** CatA requires Fe(III); CatB requires  $Mn^{2+}$  [PMID: 7500361](#). Loss of the correct metal cofactor is a mechanistic failure mode for each; CatC, by contrast, is metal-independent (a Glu general base) and is thus less vulnerable to metal limitation [PMID: 23421784](#).

**Regulatory dependency.** Expression depends on the presence of the inducer *cis,cis*-muconate and a functional regulator; without induction the pathway is silent even when substrate is present. Because the inducer is an *internal* intermediate (the CatA product), a small basal level of CatA is required to prime induction — a feed-forward dependency. In some systems TCA-cycle intermediates such as fumarate impose an additional repressive layer on the related *clc* operon, coupling aromatic catabolism to central-metabolic state [PMID: 9352923](#).

## 7. Controversies and Open Questions

**Metal identity of CatB.** A genuine point of imprecision in the literature is the divalent cation of muconate cycloisomerase. The broad enolase-superfamily framing describes a generic divalent metal, frequently written as  $Mg^{2+}$  [PMID: 8987982](#), [PMID: 19220063](#). The specific structural work on the canonical *P. putida* enzyme, however, unambiguously identifies an **essential**  $Mn^{2+}$  [PMID: 7500361](#), [PMID: 3612800](#). Reviewers should be careful not to over-generalize the superfamily's generic  $Mg^{2+}$  description to CatB; the direct structural evidence favors  $Mn^{2+}$ .

**Extent of substrate channeling / complex formation.** Whether the three catalysts physically associate into a metabolon, or simply share a cytoplasmic pool of freely diffusing intermediates, is not resolved by the current evidence. The three folds are unrelated and their oligomeric states differ (dimer, octamer, decamer), which argues against an obligate stoichiometric supercomplex, but transient channeling of the reactive, potentially toxic intermediates has not been rigorously excluded.

**Paralog function beyond *P. putida*.** The CatA/CatA2 non-redundancy is well demonstrated in *P. putida* mt-2 [PMID: 24341396](#), but whether analogous safety-valve paralogs operate in other lineages, or whether this is a *Pseudomonas*-specific adaptation tied to TOL-plasmid carriage, remains open.

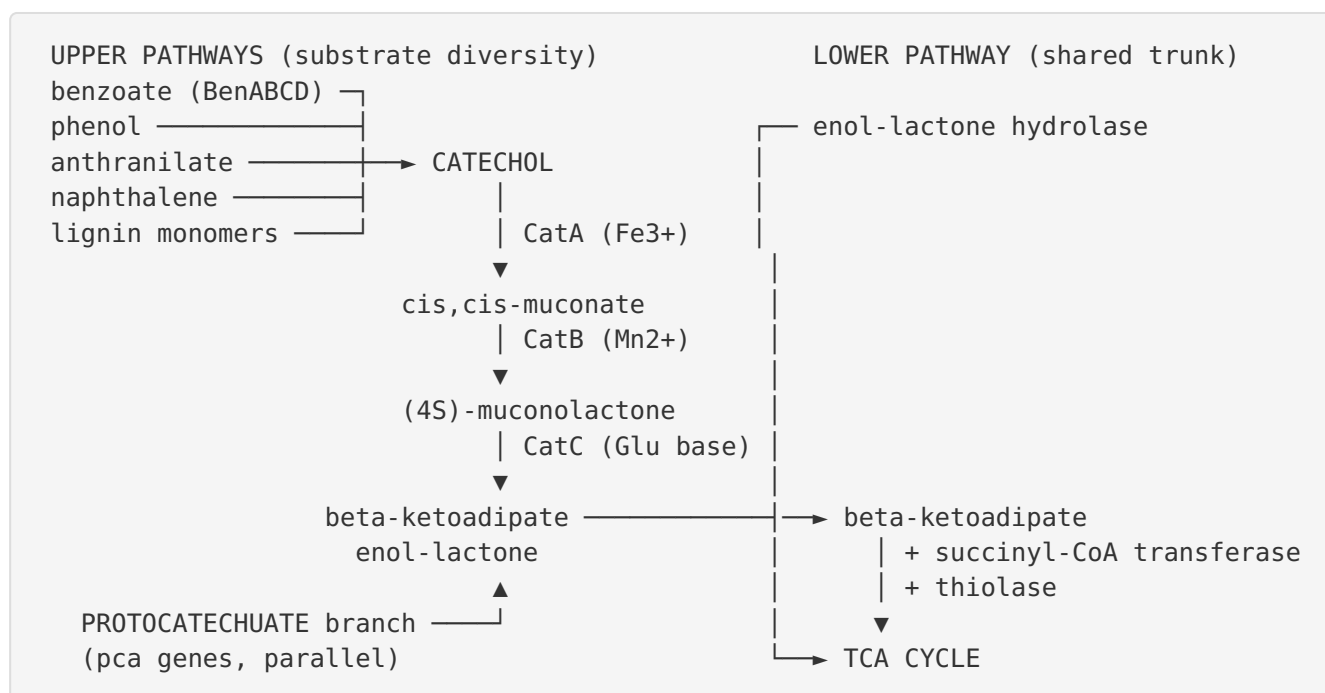
**Mechanistic transferability.** Much of the detailed active-site mechanism for the intradiol step derives from the *homologous* protocatechuate 3,4-dioxygenase rather than from catechol 1,2-dioxygenase itself [PMID: 16101286](#), [PMID: 9485360](#). Likewise, the CatC/MLI catalytic detail leans heavily on the ClcF dehalogenase homolog [PMID: 23421784](#). These extrapolations are reasonable given high structural conservation, but they are extrapolations — direct mechanistic studies on the canonical *cat*-encoded enzymes are comparatively sparse.

**Regulator diversity — convergent or divergent?** The use of LysR regulators in some lineages and IclR-type in others [PMID: 17483937](#) raises the unresolved question of whether the ancestral *cat* operon was LysR-controlled with subsequent regulator replacement, or whether regulatory recruitment happened independently in different lineages.

**Most important open questions:** 1. Is there physical channeling among CatA/CatB/CatC, and does it mitigate intermediate toxicity? 2. What determines the flux ceiling of CatA under real *ortho/meta* co-expression, and can it be engineered to prevent catechol toxicity? 3. How general is the CatA2 safety-valve strategy across aromatic-degrading bacteria? 4. What is the ancestral regulatory state of the operon?

## 8. Mechanistic Model — Synthesis

The catechol *ortho*-cleavage branch is best understood as a **three-fold funnel converging into a shared trunk**:



Three unrelated superfamilies (intradiol dioxygenase, enolase, MLI) are recruited into one operon and coordinately induced by an internal intermediate (*cis,cis*-muconate) via a lineage-variable regulator. The catalytic core is deeply conserved and ancient — ancient enough to have served as the template from which the chlorocatechol modified-*ortho* pathways were later derived. Variation accrues at the periphery: regulator identity and gene order, thermostability tuning to niche, catabolite-repression sensitivity, and paralog elaboration (CatA2). The system's principal vulnerability is the toxicity of catechol at the entry point, which explains why evolutionary reinforcement (a second dioxygenase) targets the first step specifically.

## 9. Evidence Base

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| PMID                     | Paper (abbreviated)                                                                | Role in this review                                                                                          |
|--------------------------|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| <a href="#">8905091</a>  | <i>The <math>\beta</math>-ketoadipate pathway and the biology of self-identity</i> | Authoritative review; defines the branch, convergence to $\beta$ -ketoadipate, chlorocatechol-ancestor claim |
| <a href="#">12534466</a> | <i>Genomic analysis of aromatic pathways, <i>P. putida</i> KT2440</i>              | Annotates the <i>cat/pca</i> branches and gene loci                                                          |
| <a href="#">16470664</a> | <i>Proteomic analysis of aromatic catabolism, KT2440</i>                           | Demonstrates convergence onto shared $\beta$ -ketoadipate steps                                              |
| <a href="#">10801478</a> | <i>1.8 Å structure of catechol 1,2-dioxygenase</i>                                 | CatA quaternary structure, Fe(III), helical zipper, phospholipids                                            |
| <a href="#">19300822</a> | <i>Acid-base catalysis in intradiol/extradiol dioxygenases</i>                     | Common hydroperoxide intermediate; divergent Criegee rearrangements                                          |
| <a href="#">16101286</a> | <i>Equatorial tyrosyl ligand of 3,4-PCD</i>                                        | Defines 2-His-2-Tyr Fe(III) site and tyrosinate base                                                         |
| <a href="#">9485360</a>  | <i>Axial tyrosinate in 3,4-PCD</i>                                                 | Reaction-cycle intermediates; tyrosinate role in binding/release                                             |
| <a href="#">9667935</a>  | <i>Oxygen-activating nonheme iron enzymes</i>                                      | Contrasting 2-His-1-carboxylate facial triad of extradiol enzymes                                            |
| <a href="#">10368270</a> | <i>Catechol 2,3-dioxygenase (metapyrocatechase) structure</i>                      | Defines the competing <i>meta</i> /extradiol fate                                                            |
| <a href="#">8987982</a>  | <i>The enolase superfamily</i>                                                     | Mechanistic signature CatB belongs to                                                                        |
| <a href="#">19220063</a> | <i>Stereochemically distinct MLE families</i>                                      | CatB reaction, (4S) stereochemistry, Lys catalyst, syn/anti families                                         |
| <a href="#">7500361</a>  | <i>1.85 Å structure of MLE, <i>P. putida</i> PRS2000</i>                           | Homo-octamer, ( $\beta/\alpha$ ) <sub>7</sub> $\beta$ -barrel, essential Mn <sup>2+</sup>                    |
| <a href="#">3612800</a>  | <i>Crystal structure of MLE at 3 Å</i>                                             | Independent confirmation of essential Mn <sup>2+</sup>                                                       |
| <a href="#">23376735</a> | <i>X-ray/docking of ClcF (MLI homolog)</i>                                         | Defines CatC reaction (module output)                                                                        |
| <a href="#">23421784</a> | <i>Structure/mechanism of ClcF dehalogenase</i>                                    | CatC/MLI ferredoxin fold, decamer, shared-interface Glu base                                                 |

| PMID                     | Paper (abbreviated)                                       | Role in this review                                         |
|--------------------------|-----------------------------------------------------------|-------------------------------------------------------------|
| <a href="#">22842338</a> | <i>Recombinant ClcF, mutational analysis</i>              | Confirms E27 proton abstraction, substrate-binding residues |
| <a href="#">9858745</a>  | <i>Transcriptional activation of cat/clc operons</i>      | CatR/LysR regulation, <i>cis,cis</i> -muconate inducer      |
| <a href="#">9573187</a>  | <i>CatR internal binding site</i>                         | DNA-looping repression layer                                |
| <a href="#">9171413</a>  | <i>ClcR activation of clcABD</i>                          | Conserved LysR activation mechanism in the derived pathway  |
| <a href="#">9352923</a>  | <i>Fumarate repression of clcABD</i>                      | TCA-state coupling of aromatic catabolism                   |
| <a href="#">17483937</a> | <i>catRABC operon, R. erythropolis</i>                    | IclR-type regulator, variant gene order                     |
| <a href="#">21695219</a> | <i>cat/ben genes, Halomonas organivorans</i>              | catRBCA arrangement in halophile                            |
| <a href="#">7765840</a>  | <i>Evolution of chlorocatechol pathways</i>               | Sequence relatedness of cat/chlorocatechol enzymes          |
| <a href="#">24341396</a> | <i>Second catA copy, P. putida mt-2</i>                   | CatA2 safety-valve function                                 |
| <a href="#">16478453</a> | <i>Proteome of KT2440 on different carbon sources</i>     | Catabolite repression of Cat proteins                       |
| <a href="#">23720227</a> | <i>Anthranilate degradation, cold-adapted Pseudomonas</i> | Thermolabile CatA variant                                   |
| <a href="#">17975081</a> | <i>Naphthalene toxicity, Polaromonas CJ2</i>              | Toxic-intermediate failure mode                             |

## 10. Limitations and Knowledge Gaps

- **Homolog-based mechanism.** Detailed active-site mechanisms for CatA and CatC rest substantially on close homologs (3,4-PCD and ClcF, respectively) rather than the canonical *cat*-encoded enzymes. Conservation makes the inference sound but not certain.



- **Organismal breadth.** Much of the deepest data is from *Pseudomonas*. Regulatory variation (IclR in *Rhodococcus*, LysR in *Halomonas*) is documented, but paralog and channeling questions are largely *Pseudomonas*-centric.
  - **In vitro vs. in vivo metal loading.** The  $Mn^{2+}$  assignment for CatB comes from purified/crystallized enzyme; whether metal occupancy is invariant in vivo across conditions is not established.
  - **No quantitative flux data in this synthesis.** The review integrates structural, genetic, and regulatory evidence but does not include kinetic flux measurements distinguishing rate-limiting steps under physiological loads.
  - **Channeling unresolved.** Whether the module functions as free-diffusion enzymes or a loosely organized metabolon is genuinely open.
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## 11. Proposed Follow-up Experiments / Actions

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1. **Direct metal quantification of canonical CatB across organisms** (ICP-MS on purified enzyme  $\pm$  metal-substitution kinetics) to settle the  $Mn^{2+}$ -vs- $Mg^{2+}$  question definitively for the  $\beta$ -ketoadipate-pathway cycloisomerase.
  2. **Test for physical association / channeling** among CatA, CatB, CatC using co-purification, cross-linking mass spectrometry, and single-turnover kinetics with and without downstream enzymes present, to probe intermediate-toxicity mitigation.
  3. **Comparative genomics of catA paralogs** across aromatic-degrading bacteria to determine how widespread the CatA2 safety-valve strategy is and whether it correlates with carriage of *meta*-pathway plasmids.
  4. **Direct structural/mechanistic work on the canonical cat-encoded CatA and CatC** (rather than 3,4-PCD/ClcF surrogates) to confirm transferability of the tyrosinate-base and Glu-base mechanisms.
  5. **Reconstruct the ancestral regulator** by phylogenetics of CatR (LysR) vs. IclR-type regulators across the *cat/clc/cbn* families to resolve whether regulatory control is ancestral or convergently recruited.
  6. **Systematic flux/toxicity mapping** under controlled *ortho/meta* co-expression to identify the CatA capacity threshold at which catechol becomes toxic and to guide metabolic-engineering safeguards.
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*Prepared from an autonomous 5-iteration literature synthesis; 8 findings confirmed across 37 reviewed papers. All mechanistic claims are anchored to primary structural, biochemical, genomic, or regulatory literature as cited; uncertainty is flagged explicitly where evidence relies on homologs or on single-organism data.*

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